VP, SENIOR DATA SCIENTIST

# WenTsong Shiue

## PROFILE

Wentsong Shiue, PhD. In Electrical Engineering and Computer Science
Bank of America, Vice President, Senior Data Scientist
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## SUMMARY

- With over years of experience as a data scientist, I possess a wealth of knowledge in Python, R, Matlab, and Tableau. My expertise encompasses a range of disciplines including machine learning, data mining, data visualization, statistics, and feature engineering, utilizing tools such as Numpy, Pandas, Matplotlib, Sklearn, Scipy, Seaborn, and R packages.

- I have a strong background in artificial intelligence frameworks, including the use of artificial neural networks for regression and classification problems, convolutional neural networks for image and computer vision, and recurrent neural networks for time series analysis, utilizing tools such as Keras, Theano, Tensorflow, and PyTorch.

- I have hands-on experience with a variety of statistical models, text analysis, and machine learning algorithms and methodologies, including Naive Bayes, association analysis, supervised and unsupervised learning, reinforcement learning, and more. Additionally, I have worked with AI platforms using Scala, PySpark, C++/C, Java, PostgreSQL, R, Python, Matlab, Tensorflow, Theano, PyTorch, Keras, and Tableau.

- I am well-versed in distributed processing of large data sets using tools such as Hadoop, MapReduce, Hive, and Aster, and have a deep understanding of various models in regression, classification, and clustering, such as Ridge, Lasso, ElasticNet, Decision Trees, Random Forest, Support Vector Machines, Naive Bayes, Logistics, KMeans, and ensemble methods like XGBoost, LightGBM, Bagging, AdaBoost, Gradient Boosted Machines, Extra Trees, and Random Forest. I also have expertise in feature extraction techniques such as PCA, LDA, and t-SNE.

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- I am a team player who demonstrates a willingness to both teach others and continuously learn new techniques. I work effectively under general supervision and thrive in a collaborative environment.

## EXPERIENCES

### Bank of America: Oct 2017 - Present

- Patent: Wentsong Shiue, Tony Yu Fu, and Xianhong Zhang, Number P9978-US: *Determining optimal machine learning models* will be granted by the U.S. Patent and Trademark Office on 20 Dec 2022 with Patent Number 11,531,734.

As a machine learning developer, I have led and contributed to the development of a wide range of business intelligence and data warehousing projects utilizing a full range of technologies such as Python, R, HULC, and Tableau. My work involved implementing machine learning and deep learning algorithms on large datasets, working closely with line-of-business and MRM validators for performance analysis and bias testing, collaborating with data analysts and QA specialists, and providing support to numerous analytical teams. I have established efficient processes, performance tools, and best development practices to ensure rapid and high-quality project delivery. Additionally, I have introduced technical documentation standards, including requirements, workflow diagrams, and code.

**Model Developer (CAML, Device, Wire, ACH, BillPay):**

I have developed binary classification models using XGBoost and LightGBM for CAML, Wire, ACH, and Bill Pay applications. These models are designed to perform binary classification tasks and are built using cutting-edge machine learning techniques. The use of XGBoost and LightGBM provides improved performance and accuracy in these applications, making them better equipped to handle complex data and provide insightful results.

**Bias Testing (P2P, Debit, Wire, ACH, Bill Pay)**

The assessment of biases in machine learning models is critical for every company working in this field. I am honored to have been selected to work on a project focused on testing for biases based on factors such as race, gender, age, and marital status. The models will undergo statistical tests to measure recall, precision, false positive rate, and other key metrics, ensuring that they are fair and unbiased in their predictions. This is

an important responsibility and I am committed to delivering accurate results that uphold the integrity and quality of our company's work in this field.

**Fraud Detection: (CAML, Credit, P2P, Debit, Device, Wire, ACH, Bill Pay)**

As a lead member of the model team, I am involved in a variety of machine learning projects on a daily basis. Our processes involve defining the machine learning problem and features, retrieving raw data from HULC, encoding the data, calculating correlation coefficients and filtering out high-correlation features, and selecting features through techniques such as removing single-value columns, zero feature importance, and high-correlation features. We also fine-tune our model parameters using optimization techniques such as random search, optuna, and Bayesian optimization. After generating the model, we evaluate its performance using metrics such as recall, precision, f1, roc_auc, and ROC/PRC curves, and then deploy it for production. We use Splunk for error analysis and interpretable machine learning platforms such as PDP, SHAP, and ELI to explain the black-box model.

One of the challenges in complex machine learning models is transparency; it can be difficult to understand why a model is making a certain prediction and which features are contributing to it. We use techniques such as SHAP (SHapley Additive exPlanation), LIME (Local Interpretable Model-agnostic Explanations), and Eli5 to explain the predictions of the black-box model. The SHAP approach involves computing the contribution of each feature to a prediction and visualizing it through force plots, summary plots, positive/negative effects, SHAP importance, dependence plots, and decision plots. LIME is a local surrogate model that uses linear regression or decision tree models to explain predictions at the local boundary. ELI5 helps to debug machine learning classifiers and explain their top predictions in a visually appealing and easy-to-understand manner.

### Uniqlinc, Inc. June 2013 - Oct 2017, Director of Data Science

As a member of the BI team, I was responsible for supporting numerous machine learning and analytical teams in delivering high-quality reports to thousands of employees. I played a key role in establishing efficient processes, tools, and development practices to ensure rapid delivery of projects. Additionally, I introduced standards for technical documentation, including the creation of detailed requirements, workflow diagrams, and code support. To further streamline the development process, I implemented a test-driven project development approach using Git. Additionally, I was tasked with managing a custom, highly scalable, and performant Python ETL application.

The following technologies are utilized in detail: Artificial Neural Network and Convolutional Neural Network. Dimensionality Reduction techniques include Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), and kernel PCA. Model selection and boosting methods such as K-fold Cross Validation and XGBoost are employed. Reinforcement learning algorithms, including Upper Confidence Bound (UCB) and Thomson Sampling, are also utilized.

### Alvaview, Inc.: June 2006 - June 2013, Chief Data Scientist
### Data Science Division

Responsibilities: As a lead in the team, I have made significant contributions to the development of various Business Intelligence, data warehousing and reporting projects using a full spectrum of technologies including Python, R, Matlab, PostgreSQL, and Tableau. The projects are based on the applications of Machine Learning and Deep Learning for big data sets. I have managed a team consisting of developers, data analysts and QA specialists to deliver high-quality projects.

Technology Used: The following is a detailed list of the technologies used in the projects: Regression models for continuous dependent variables include Simple Linear Regression, Multiple Linear Regression with backward elimination, Polynomial Regression, Support Vector Regression, Decision Tree Regression, and Random Forest Regression. Classification models for categorical dependent variables include Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Kernel SVM, Naive Bayes, Decision Tree / Random Forest Classification. Clustering models for data without dependent variables include K-Means Clustering and Hierarchical Clustering. Association Rule Learning algorithms used are Apriori and Eclat.

### FreeScale Semiconductor: Jan 2004 – June 2006, Austin TX System, Senior Data Modeling Engineer

Multimedia Application Division Patent: *A Novel Table-less Modeling Solution for Dynamic Process Temperature Compensation (DPTV) System using pseudo regression method*.

In the area of system data modeling, I focused on analyzing the power consumption of i.MX chipset, examining system bus data bandwidth and video power, as well as modeling and analyzing system performance data. Additionally, I investigated power management and memory power, taking into consideration the impact of data accesses. My work also involved analyzing the i.MX architecture bus and memory bandwidth.

## EDUCATION:

### Arizona State University (ASU): Aug 1996 - Jun 2000, Tempe AZ,

PhD in Electrical Engineering and Computer Science with a Dissertation focused on "*Memory Data Modeling and Exploration in Low Power Embedded Systems.*" The research on cache data power modeling, presented in the dissertation, has been implemented in the SimplePower tool at Penn State University (PSU), a cycle-accurate energy estimation tool.

### Western Michigan University (WMU): Jan 1989 - Jun 1991, Kalamazoo MI,

M.S. in Electrical Engineering and Computer Engineering, with a thesis titled "*Image Data Processing using Discrete Cosine Transform and Huffman Coding.*" The research focused on utilizing these techniques for image data processing.

### Ming Chi University of Technology (MCUT): Jun 1980 - Jun 1985, Taiwan,

B.S. in Electrical Engineering with a thesis on "I*ntelligent Machine Learning for Robotic Motion Control*." The research explored the application of machine learning for controlling the motion of robots.

## PUBLICATIONS

Research publications can be found on Microsoft Academic Search and Google Scholar (https://scholar.google.com/scholar?hl=zh-TW&q=wen-tsong+shiue&lr=&oq=).

## HONORS

Achieved 3rd rank out of 120,000 students in the Taiwan National College Examination in 1979.